Z-transform

PROBLEMS

3.1 Determine the z-transform of the following signals.

(a) $x(n) = \{3, 0, 0, 0, 0, \underset{\uparrow}{6}, 1, -4\}$

(b) $x(n) = \begin{cases} (\frac{1}{2})^n, & n \geq 5 \\ 0, & n \leq 4 \end{cases}$

3.2 Determine the z-transforms of the following signals and sketch the corresponding pole–zero patterns.

(a) $x(n) = (1 + n)u(n)$

(b) $x(n) = (a^n + a^{-n})u(n)$, a real

(c) $x(n) = (-1)^n 2^{-n} u(n)$

(d) $x(n) = (na^n \sin \omega_0 n)u(n)$

(e) $x(n) = (na^n \cos \omega_0 n)u(n)$

(f) $x(n) = Ar^n \cos(\omega_0 n + \phi)u(n)$, $0 < r < 1$

(g) $x(n) = \frac{1}{2}(n^2 + n)(\frac{1}{3})^{n-1} u(n-1)$

(h) $x(n) = (\frac{1}{2})^n [u(n) - u(n-10)]$

3.3 Determine the z-transforms and sketch the ROC of the following signals.

(a) $x_1(n) = \begin{cases} (\frac{1}{3})^n, & n \geq 0 \\ (\frac{1}{2})^{-n}, & n < 0 \end{cases}$

(b) $x_2(n) = \begin{cases} (\frac{1}{3})^n - 2^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$

(c) $x_3(n) = x_1(n + 4)$

(d) $x_4(n) = x_1(-n)$

3.4 Determine the z-transform of the following signals.

(a) $x(n) = n(-1)^n u(n)$

(b) $x(n) = n^2 u(n)$

(c) $x(n) = -na^n u(-n-1)$

(d) $x(n) = (-1)^n \left(\cos \frac{\pi}{3} n\right) u(n)$

(e) $x(n) = (-1)^n u(n)$

(f) $x(n) = \{\underset{\uparrow}{1}, 0, -1, 0, 1, -1, \ldots\}$

3.5 Determine the regions of convergence of right-sided, left-sided, and finite-duration two-sided sequences.

3.53 A causal pole–zero system is BIBO stable if its poles are inside the unit circle. Consider now a pole–zero system that is BIBO stable and has its poles inside the unit circle. Is the system always causal? [*Hint:* Consider the systems $h_1(n) = a^n u(n)$ and $h_2(n) = a^n u(n+3)$, $|a| < 1$.]

DTFT

4.9 Compute the Fourier transform of the following signals.

(a) $x(n) = u(n) - u(n-6)$

(b) $x(n) = 2^n u(-n)$

(c) $x(n) = (\frac{1}{4})^n u(n+4)$

(d) $x(n) = (\alpha^n \sin \omega_0 n) u(n) \qquad |\alpha| < 1$

(e) $x(n) = |\alpha|^n \sin \omega_0 n \qquad |\alpha| < 1$

(f) $x(n) = \begin{cases} 2 - (\frac{1}{2})n, & |n| \le 4 \\ 0, & \text{elsewhere} \end{cases}$

(g) $x(n) = \{-2, -1, \underset{\uparrow}{0}, 1, 2\}$

(h) $x(n) = \begin{cases} A(2M + 1 - |n|), & |n| \le M \\ 0, & |n| > M \end{cases}$

Sketch the magnitude and phase spectra for parts (a), (f), and (g).

4.19 Let $x(n)$ be a signal with Fourier transform as shown in Fig. P4.19. Determine and sketch the Fourier transforms of the following signals.

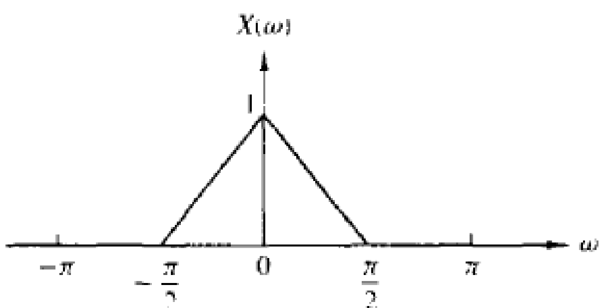

Figure P4.19

(a) $x_1(n) = x(n)\cos(\pi n/4)$ **(b)** $x_2(n) = x(n)\sin(\pi n/2)$

(c) $x_3(n) = x(n)\cos(\pi n/2)$ **(d)** $x_4(n) = x(n)\cos \pi n$

Note that these signal sequences are obtained by *amplitude modulation* of a carrier $\cos \omega_c n$ or $\sin \omega_c n$ by the sequence $x(n)$.

DFT

5.17* Determine the eight-point DFT of the signal

$$x(n) = \{1, 1, 1, 1, 1, 1, 0, 0\}$$

and sketch its magnitude and phase.

5.18 A linear time-invariant system with frequency response $H(\omega)$ is excited with the periodic input

$$x(n) = \sum_{k=-\infty}^{\infty} \delta(n - kN)$$

Suppose that we compute the N-point DFT $Y(k)$ of the samples $y(n)$, $0 \le n \le N - 1$ of the output sequence. How is $Y(k)$ related to $H(\omega)$?

5.21 Let $x_a(t)$ be an analog signal with bandwidth $B = 3$ kHz. We wish to use a $N = 2^m$-point DFT to compute the spectrum of the signal with a resolution less than or equal to 50 Hz. Determine (a) the minimum sampling rate, (b) the minimum number of required samples, and (c) the minimum length of the analog signal record.

5.23 Compute the N-point DFTs of the signals

(a) $x(n) = \delta(n)$

(b) $x(n) = \delta(n - n_0) \quad 0 < n_0 < N$

(c) $x(n) = a^n \quad 0 \le n \le N - 1$

(d) $x(n) = \begin{cases} 1, & 0 \le n \le N/2 - 1 (N \text{ even}) \\ 0, & N/2 \le n \le N - 1 \end{cases}$

(e) $x(n) = e^{j(2\pi/N)k_0} \quad 0 \le n \le N - 1$

5.25 **(a)** Determine the Fourier transform $X(\omega)$ of the signal

$$x(n) = \{1, 2, \underset{\uparrow}{3}, 2, 1, 0\}$$

(b) Compute the 6-point DFT $V(k)$ of the signal

$$v(n) = \{3, 2, 1, 0, 1, 2\}$$

(c) Is there any relation between $X(\omega)$ and $V(k)$? Explain.